JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Differentiation

JEE Main Mathematics · Previous Year Questions with Answer Key

82	73	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	69 questions
Numerical	13 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 69 questions

Q1. JEE Main 2026 (Online) 4th April Morning Shift

If $y = \tan^{-1}\left(\frac{3\cos x - 4\sin x}{4\cos x + 3\sin x}\right) + 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{3}}{2}$ is equal to :

(A) 3 (B) -1 (C) 1 (D) 2

Q2. JEE Main 2026 (Online) 21st January Evening Shift

Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3), x \in \mathbb{R}$. Then the value of $f'(5)$ is :

(A) $\frac{657}{5}$ (B) $\frac{117}{5}$ (C) $\frac{2}{5}$ (D) $\frac{62}{5}$

Q3. JEE Main 2025 (Online) 3rd April Morning Shift

If $y(x) = \begin{vmatrix} \sin x & \cos x & \sin x + \cos x + 1 \\ 27 & 28 & 27 \\ 1 & 1 & 1 \end{vmatrix}, x \in \mathbb{R}$, then $\frac{d^2y}{dx^2} + y$ is equal to

(A) 28 (B) 27 (C) -1 (D) 1

Q4. JEE Main 2025 (Online) 2nd April Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $(\sin x \cos y)(f(2x+2y) - f(2x-2y)) = (\cos x \sin y)(f(2x+2y) + f(2x-2y))$, for all $x, y \in \mathbf{R}$. If $f'(0) = \frac{1}{2}$, then the value of $24 f''\left(\frac{5\pi}{3}\right)$ is :

(A) 2 (B) 3 (C) -3 (D) -2

Q5. JEE Main 2025 (Online) 24th January Evening Shift

Let $f : (0, \infty) \to \mathbf{R}$ be a function which is differentiable at all points of its domain and satisfies the condition $x^2 f'(x) = 2x f(x) + 3$, with $f(1) = 4$. Then $2f(2)$ is equal to :

(A) 19 (B) 23 (C) 29 (D) 39

Q6. JEE Main 2024 (Online) 9th April Evening Shift

If $\log_e y = 3\sin^{-1} x$, then $(1 - x^2)y'' - xy'$ at $x = \frac{1}{2}$ is equal to

(A) $9e^{\pi/2}$ (B) $9e^{\pi/6}$ (C) $3e^{\pi/2}$ (D) $3e^{\pi/6}$

Q7. JEE Main 2024 (Online) 9th April Morning Shift

Let $f(x) = ax^3 + bx^2 + cx + 41$ be such that $f(1) = 40$, $f'(1) = 2$ and $f''(1) = 4$. Then $a^2 + b^2 + c^2$ is equal to:

(A) 54 (B) 51 (C) 73 (D) 62

Q8. JEE Main 2024 (Online) 6th April Evening Shift

Suppose for a differentiable function $h, h(0)=0, h(1)=1$ and $h'(0)=h'(1)=2$. If $g(x)=h\left(\mathrm{e}^{x}\right)\mathrm{e}^{h(x)}$, then $g'(0)$ is equal to:

(A) 4 (B) 5 (C) 3 (D) 8

Q9. JEE Main 2024 (Online) 6th April Morning Shift

If $f(x)=\begin{cases}x^3\sin\left(\frac{1}{x}\right), & x\neq 0\\ 0 & ,x=0\end{cases}$, then

(A) $f''(0)=0$ (B) $f''(0)=1$

(C) $f''\left(\frac{2}{\pi}\right)=\frac{24-\pi^2}{2\pi}$ (D) $f''\left(\frac{2}{\pi}\right)=\frac{12-\pi^2}{2\pi}$

Q10. JEE Main 2024 (Online) 6th April Morning Shift

Let $f:(-\infty,\infty)-\{0\}\to\mathbb{R}$ be a differentiable function such that $f'(1)=\lim\limits_{a\to\infty}a^2 f\left(\frac{1}{a}\right)$. Then $\lim\limits_{a\to\infty}\frac{a(a+1)}{2}\tan^{-1}\left(\frac{1}{a}\right)+a^2-2\log_e a$ is equal to

(A) $\frac{5}{2}+\frac{\pi}{8}$ (B) $\frac{3}{8}+\frac{\pi}{4}$ (C) $\frac{3}{4}+\frac{\pi}{8}$ (D) $\frac{3}{2}+\frac{\pi}{4}$

Q11. JEE Main 2024 (Online) 5th April Evening Shift

If $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$, then at $\theta=\frac{\pi}{2}$, $y''+y'+y$ is equal to :

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

Q12. JEE Main 2024 (Online) 4th April Morning Shift

Let $f(x)=x^5+2\mathrm{e}^{x/4}$ for all $x\in\mathbf{R}$. Consider a function $g(x)$ such that $(g\circ f)(x)=x$ for all $x\in\mathbf{R}$. Then the value of $8g'(2)$ is :

(A) 4 (B) 2 (C) 16 (D) 8

Q13. JEE Main 2024 (Online) 31st January Morning Shift

If $f(x)=\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}$ for all $x\in\mathbb{R}$, then $2f(0)+f'(0)$ is equal to

(A) 24 (B) 18 (C) 42 (D) 48

Q14. JEE Main 2024 (Online) 30th January Evening Shift

Let $f:\mathbb{R}-\{0\}\to\mathbb{R}$ be a function satisfying $f\left(\frac{x}{y}\right)=\frac{f(x)}{f(y)}$ for all $x, y, f(y)\neq 0$. If $f'(1)=2024$, then

(A) $xf'(x)+2024f(x)=0$ (B) $xf'(x)-2023f(x)=0$

(C) $xf'(x)-2024f(x)=0$ (D) $xf'(x)+f(x)=2024$

Q15. JEE Main 2024 (Online) 30th January Morning Shift

Let $g : \mathbf{R} \to \mathbf{R}$ be a non constant twice differentiable function such that $g'\left(\frac{1}{2}\right) = g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x) = \frac{1}{2}[g(x) + g(2-x)]$, then

(A) $f''(x) = 0$ for at least two x in $(0, 2)$

(B) $f'\left(\frac{3}{2}\right) + f'\left(\frac{1}{2}\right) = 1$

(C) $f''(x) = 0$ for no x in $(0, 1)$

(D) $f''(x) = 0$ for exactly one x in $(0, 1)$

Q16. JEE Main 2024 (Online) 30th January Morning Shift

If $f(x) = \begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$, then $\frac{1}{5}f'(0)$ is equal to :

(A) 2 (B) 1 (C) 0 (D) 6

Q17. JEE Main 2024 (Online) 29th January Evening Shift

Let $y = \log_e\left(\frac{1-x^2}{1+x^2}\right), -1 < x < 1$. Then at $x = \frac{1}{2}$, the value of $225\,(y' - y'')$ is equal to

(A) 732 (B) 736 (C) 742 (D) 746

Q18. JEE Main 2024 (Online) 29th January Morning Shift

Suppose $f(x) = \frac{(2^x+2^{-x})\tan x\sqrt{\tan^{-1}(x^2-x+1)}}{(7x^2+3x+1)^3}$. Then the value of $f'(0)$ is equal to

(A) π (B) $\sqrt{\pi}$ (C) 0 (D) $\frac{\pi}{2}$

Q19. JEE Main 2023 (Online) 13th April Morning Shift

For the differentiable function $f : \mathbb{R} - \{0\} \to \mathbb{R}$, let $3f(x) + 2f\left(\frac{1}{x}\right) = \frac{1}{x} - 10$, then $\left|f(3) + f'\left(\frac{1}{4}\right)\right|$ is equal to

(A) 13 (B) $\frac{29}{5}$ (C) $\frac{33}{5}$ (D) 7

Q20. JEE Main 2023 (Online) 8th April Morning Shift

Let $f(x) = \frac{\sin x+\cos x-\sqrt{2}}{\sin x-\cos x}, x \in [0, \pi] - \left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to

(A) $\frac{2}{3\sqrt{3}}$ (B) $\frac{2}{9}$ (C) $\frac{-1}{3\sqrt{3}}$ (D) $\frac{-2}{3}$

Q21. JEE Main 2023 (Online) 6th April Morning Shift

If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at $(2, 2)$ is equal to :

(A) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$ (B) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$

(C) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$ (D) $-\left(\frac{3+\log_e 4}{2+\log_e 8}\right)$

Q22. JEE Main 2023 (Online) 1st February Evening Shift

If $y(x) = x^x, x > 0$, then $y''(2) - 2y'(2)$ is equal to

(A) $4(\log_e 2)^2 + 2$ (B) $8\log_e 2 - 2$ (C) $4\log_e 2 + 2$ (D) $4(\log_e 2)^2 - 2$

Q23. JEE Main 2023 (Online) 1st February Morning Shift

Let $f(x) = 2x + \tan^{-1}x$ and $g(x) = \log_e(\sqrt{1+x^2} + x), x \in [0, 3]$. Then

(A) there exists $\widehat{x} \in [0, 3]$ such that $f'(\widehat{x}) < g'(\widehat{x})$

(B) there exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x), \forall x \in (x_1, x_2)$

(C) $\min f'(x) = 1 + \max g'(x)$

(D) $\max f(x) > \max g(x)$

Q24. JEE Main 2023 (Online) 31st January Morning Shift

Let $y = f(x) = \sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3 + 5x^2 + 1\right)^{\frac{3}{2}}\right)\right)\right)$. Then, at x = 1,

(A) $2y' + \sqrt{3}\pi^2 y = 0$ (B) $y' + 3\pi^2 y = 0$

(C) $\sqrt{2}y' - 3\pi^2 y = 0$ (D) $2y' + 3\pi^2 y = 0$

Q25. JEE Main 2023 (Online) 29th January Evening Shift

Let f and g be the twice differentiable functions on $\mathbb{R}$ such that

$f''(x) = g''(x) + 6x$

$f'(1) = 4g'(1) - 3 = 9$

$f(2) = 3g(2) = 12.$

Then which of the following is NOT true?

(A) $g(-2) - f(-2) = 20$

(B) There exists $x_0 \in (1, 3/2)$ such that $f(x_0) = g(x_0)$

(C) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

(D) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

Q26. JEE Main 2023 (Online) 25th January Morning Shift

Let $y(x) = (1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y' - y''$ at $x = -1$ is equal to

(A) 496 (B) 976 (C) 464 (D) 944

Q27. JEE Main 2023 (Online) 24th January Evening Shift

If $f(x) = x^3 - x^2 f'(1) + xf''(2) - f'''(3), x \in \mathbb{R}$, then

(A) $2f(0) - f(1) + f(3) = f(2)$ (B) $f(1) + f(2) + f(3) = f(0)$

(C) $f(3) - f(2) = f(1)$ (D) $3f(1) + f(2) = f(3)$

Q28. JEE Main 2022 (Online) 28th July Evening Shift

Let $x(t) = 2\sqrt{2}\cos t\sqrt{\sin 2t}$ and

$y(t) = 2\sqrt{2}\sin t\sqrt{\sin 2t}, t \in \left(0, \frac{\pi}{2}\right)$.

Then $\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$ at $t = \frac{\pi}{4}$ is equal to :

(A) $\frac{-2\sqrt{2}}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{-2}{3}$

Q29. JEE Main 2022 (Online) 26th July Evening Shift

The value of $\log_e 2\frac{d}{dx}(\log_{\cos x}\operatorname{cosec} x)$ at $x = \frac{\pi}{4}$ is

(A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) -4 (D) 4

Q30. JEE Main 2022 (Online) 27th June Morning Shift

If $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$, then :

(A) $x^2y'' + xy' - 25y = 0$ (B) $x^2y'' - xy' - 25y = 0$

(C) $x^2y'' - xy' + 25y = 0$ (D) $x^2y'' + xy' + 25y = 0$

Q31. JEE Main 2022 (Online) 25th June Morning Shift

Let f : R $\to$ R be defined as $f(x) = x^3 + x - 5$. If g(x) is a function such that $f(g(x)) = x, \forall' x' \in R$, then g'(63) is equal to _____.

(A) $\frac{1}{49}$ (B) $\frac{3}{49}$ (C) $\frac{43}{49}$ (D) $\frac{91}{49}$

Q32. JEE Main 2022 (Online) 24th June Evening Shift

If $y = \tan^{-1}\left(\sec x^3 - \tan x^3\right)$, $\frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then

(A) $xy'' + 2y' = 0$ (B) $x^2y'' - 6y + \frac{3\pi}{2} = 0$

(C) $x^2y'' - 6y + 3\pi = 0$ (D) $xy'' - 4y' = 0$

Q33. JEE Main 2021 (Online) 27th August Evening Shift

If $y(x) = \cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right), x \in \left(\frac{\pi}{2}, \pi\right)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{6}$ is :

(A) $-\frac{1}{2}$ (B) -1 (C) $\frac{1}{2}$ (D) 0

Q34. JEE Main 2021 (Online) 26th August Morning Shift

Let $f(x) = \cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right)$, 0 < x < 1. Then :

(A) $(1-x)^2f'(x) - 2(f(x))^2 = 0$ (B) $(1+x)^2f'(x) + 2(f(x))^2 = 0$

(C) $(1-x)^2f'(x) + 2(f(x))^2 = 0$ (D) $(1+x)^2f'(x) - 2(f(x))^2 = 0$

Q35. JEE Main 2020 (Online) 5th September Evening Slot

The derivative of $\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ with respect to $\tan^{-1}\left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ at x = $\frac{1}{2}$ is :

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{2\sqrt{3}}{5}$ (C) $\frac{\sqrt{3}}{10}$ (D) $\frac{\sqrt{3}}{12}$

Q36. JEE Main 2020 (Online) 4th September Morning Slot

If $\left(a+\sqrt{2}b\cos x\right)\left(a-\sqrt{2}b\cos y\right)=a^2-b^2$

where a > b > 0, then $\frac{dx}{dy}$ $at\left(\frac{\pi}{4},\frac{\pi}{4}\right)$ is :

(A) $\frac{a-2b}{a+2b}$ (B) $\frac{a-b}{a+b}$ (C) $\frac{a+b}{a-b}$ (D) $\frac{2a+b}{2a-b}$

Q37. JEE Main 2020 (Online) 3rd September Morning Slot

If $y^2 + \log_e(\cos^2 x) = y$, $x \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$, then :

(A) |y"(0)| = 2 (B) |y'(0)| + |y"(0)| = 3

(C) y"(0) = 0 (D) |y'(0)| + |y"(0)| = 1

Q38. JEE Main 2020 (Online) 9th January Evening Slot

If $x = 2\sin\theta - \sin 2\theta$ and $y = 2\cos\theta - \cos 2\theta$, $\theta \in [0, 2\pi]$, then $\frac{d^2y}{dx^2}$ at $\theta = \pi$ is :

(A) $\frac{3}{8}$ (B) $\frac{3}{2}$ (C) $\frac{3}{4}$ (D) $-\frac{3}{4}$

Q39. JEE Main 2020 (Online) 9th January Evening Slot

Let f and g be differentiable functions on R such that fog is the identity function. If for some a, b $\in$ R, g'(a) = 5 and g(a) = b, then f'(b) is equal to :

(A) 1 (B) 5 (C) $\frac{2}{5}$ (D) $\frac{1}{5}$

Q40. JEE Main 2020 (Online) 8th January Morning Slot

Let $f(x) = (\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1$, |x| > 1.
If $\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}\left(\sin^{-1}(f(x))\right)$ and $y\left(\sqrt{3}\right) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to :

(A) $\frac{5\pi}{6}$ (B) $-\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

Q41. JEE Main 2020 (Online) 7th January Evening Slot

Let y = y(x) be a function of x satisfying

$y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and

$y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at x = $\frac{1}{2}$, is equal to :

(A) $\frac{2}{\sqrt{5}}$ (B) $-\frac{\sqrt{5}}{2}$ (C) $\frac{\sqrt{5}}{2}$ (D) $-\frac{\sqrt{5}}{4}$

Q42. JEE Main 2020 (Online) 7th January Morning Slot

Let $x^k + y^k = a^k$, (a, k > 0) and $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then k is:

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{4}{3}$ (D) $\frac{3}{2}$

Q43. JEE Main 2020 (Online) 7th January Morning Slot

If $y(\alpha) = \sqrt{2\left(\frac{\tan\alpha+\cot\alpha}{1+\tan^2\alpha}\right) + \frac{1}{\sin^2\alpha}}, \alpha \in \left(\frac{3\pi}{4}, \pi\right)$

$\frac{dy}{d\alpha}\ at\ \alpha = \frac{5\pi}{6} is$:

(A) 4 (B) -4 (C) $\frac{4}{3}$ (D) $-\frac{1}{4}$

Q44. JEE Main 2019 (Online) 12th April Evening Slot

The derivative of $\tan^{-1}\left(\frac{\sin x-\cos x}{\sin x+\cos x}\right)$, with respect to $\frac{x}{2}$, where $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is :

(A) 1 (B) 2 (C) $\frac{2}{3}$ (D) $\frac{1}{2}$

Q45. JEE Main 2019 (Online) 12th April Morning Slot

If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at x = 0 is equal to :

(A) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (B) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$

(C) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$ (D) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$

Q46. JEE Main 2019 (Online) 10th April Evening Slot

Let f(x) = $\log_e$(sin x), (0 < x < π) and g(x) = $\sin^{-1}(e^{-x})$, ($x \geq 0$). If α is a positive real number such that a = (fog)'(α) and b = (fog)(α), then :

(A) $a\alpha^2 + b\alpha - a = -2\alpha^2$ (B) $a\alpha^2 + b\alpha + a = 0$

(C) $a\alpha^2 - b\alpha - a = 0$ (D) $a\alpha^2 - b\alpha - a = 1$

Q47. JEE Main 2019 (Online) 8th April Evening Slot

If $f(1) = 1, f'(1) = 3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$ is :

(A) 33 (B) 12 (C) 9 (D) 15

Q48. JEE Main 2019 (Online) 8th April Morning Slot

If $2y = \left(\cot^{-1}\left(\frac{\sqrt{3}\cos x+\sin x}{\cos x-\sqrt{3}\sin x}\right)\right)^2$,

$x \in \left(0, \frac{\pi}{2}\right)$ then $\frac{dy}{dx}$ is equal to:

(A) $2x - \frac{\pi}{3}$ (B) $\frac{\pi}{6} - x$ (C) $\frac{\pi}{3} - x$ (D) $x - \frac{\pi}{6}$

Q49. JEE Main 2019 (Online) 12th January Morning Slot

For $x > 1$, if $(2x)^{2y} = 4e^{2x-2y}$,

then $(1 + \log_e 2x)^2 \frac{dy}{dx}$ is equal to :

(A) $\frac{x\log_e 2x-\log_e 2}{x}$ (B) $\log_e 2x$

(C) $x \log_e 2x$ (D) $\frac{x\log_e 2x+\log_e 2}{x}$

Q50. JEE Main 2019 (Online) 11th January Morning Slot

If $x\log_e(\log_e x) - x^2 + y^2 = 4 (y > 0)$, then $\frac{dy}{dx}$ at $x = e$ is equal to :

(A) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (B) $\frac{(1+2e)}{\sqrt{4+e^2}}$

(C) $\frac{(2e-1)}{2\sqrt{4+e^2}}$ (D) $\frac{e}{\sqrt{4+e^2}}$

Q51. JEE Main 2019 (Online) 10th January Morning Slot

Let $f : R \to R$ be a function such that $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3), x \in R$. Then $f(2)$ equals -

(A) 30 (B) -2 (C) -4 (D) 8

Q52. JEE Main 2019 (Online) 9th January Evening Slot

If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is :

(A) $\frac{1}{3\sqrt{2}}$ (B) $\frac{1}{6\sqrt{2}}$ (C) $\frac{3}{2\sqrt{2}}$ (D) $\frac{1}{6}$

Q53. JEE Main 2018 (Online) 16th April Morning Slot

If $x = \sqrt{2^{\cos ec^{-1}t}}$ and $y = \sqrt{2^{sec^{-1}t}}$ $(|t| \geq 1)$, then $\frac{dy}{dx}$ is equal to :

(A) $\frac{y}{x}$ (B) $\frac{x}{y}$ (C) $-\frac{y}{x}$ (D) $-\frac{x}{y}$

Q54. JEE Main 2018 (Online) 15th April Evening Slot

If f(x) = $\sin^{-1}\left(\frac{2\times 3^x}{1+9^x}\right)$, then f'$\left(-\frac{1}{2}\right)$ equals :

(A) $-\sqrt{3}\log_e\sqrt{3}$ (B) $\sqrt{3}\log_e\sqrt{3}$ (C) $-\sqrt{3}\log_e 3$ (D) $\sqrt{3}\log_e 3$

Q55. JEE Main 2018 (Online) 15th April Morning Slot

If $f(x)=\begin{vmatrix}\cos x & x & 1\\ 2\sin x & x^2 & 2x\\ \tan x & x & 1\end{vmatrix}$, then $\lim\limits_{x\to 0}\frac{f'(x)}{x}$

(A) does not exist. (B) exists and is equal to 2.

(C) exists and is equal to 0. (D) exists and is equal to $-$ 2.

Q56. JEE Main 2018 (Online) 15th April Morning Slot

If $x^2 + y^2 + \sin y = 4$, then the value of $\frac{d^2y}{dx^2}$ at the point $(-2,0)$ is :

(A) $-$ 34 (B) $-$ 32 (C) 4 (D) $-$ 2

Q57. JEE Main 2017 (Online) 9th April Morning Slot

Let f be a polynomial function such that

f (3x) = f ' (x) . f " (x), for all x $\in$ **R**. Then :

(A) f (2) + f ' (2) = 28 (B) f " (2) $-$ f ' (2) = 0

(C) f " (2) $-$ f (2) = 4 (D) f (2) $-$ f ' (2) + f " (2) = 10

Q58. JEE Main 2017 (Online) 8th April Morning Slot

If y = $\left[x+\sqrt{x^2-1}\right]^{15}+\left[x-\sqrt{x^2-1}\right]^{15}$,

then $(x^2-1)\frac{d^2y}{dx^2}+x\frac{dy}{dx}$ is equal to :

(A) 125 y (B) 124 y^2 (C) 225 y^2 (D) 225 y

Q59. JEE Main 2017 (Offline)

If for $x\in\left(0,\frac{1}{4}\right)$, the derivatives of

$\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x}.g(x)$, then $g(x)$ equals

(A) $\frac{3x\sqrt{x}}{1-9x^3}$ (B) $\frac{3x}{1-9x^3}$

(C) $\frac{3}{1+9x^3}$ (D) $\frac{9}{1+9x^3}$

Q60. JEE Main 2014 (Offline)

If g is the inverse of a function f and $f'(x) = \frac{1}{1+x^5}$, then $g'(x)$ is equal to:

(A) $\frac{1}{1+\{g(x)\}^5}$ (B) $1+\{g(x)\}^5$

(C) $1+x^5$ (D) $5x^4$

Q61. JEE Main 2013 (Offline)

If $y = \sec\left(\tan^{-1}x\right)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 1 (D) $\sqrt{2}$

Q62. AIEEE 2011

$\frac{d^2x}{dy^2}$ equals:

(A) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$

(B) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$

(C) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$

(D) $\left(\frac{d^2y}{dx^2}\right)^{-1}$

Q63. AIEEE 2010

Let $f:(-1,1)\to R$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x)+2)]^2$. Then $g'(0) =$

(A) -4 (B) 0 (C) -2 (D) 4

Q64. AIEEE 2009

Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals

(A) 1 (B) $\log 2$ (C) $-\log 2$ (D) -1

Q65. AIEEE 2006

If $x^m.y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is

(A) $\frac{y}{x}$ (B) $\frac{x+y}{xy}$ (C) xy (D) $\frac{x}{y}$

Q66. AIEEE 2004

If $x = e^{y+e^{y+\ldots\ldots\infty}}$, $x>0$, then $\frac{dy}{dx}$ is

(A) $\frac{1+x}{x}$ (B) $\frac{1}{x}$ (C) $\frac{1-x}{x}$ (D) $\frac{x}{1+x}$

Q67. AIEEE 2003

If $f(x) = x^n$, then the value of
$f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \ldots\ldots\ldots \frac{(-1)^n f^n(1)}{n!}$ is

(A) 1 (B) 2^n (C) $2^n - 1$ (D) 0

Q68. AIEEE 2003

Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in $A.P$, then $f'(a), f'(b), f'(c)$ are in

(A) Arithmetic -Geometric Progression (B) $A.P$

(C) $G.P$ (D) $H.P$

Q69. AIEEE 2002

If $y = \left(x + \sqrt{1+x^2}\right)^n$, then $\left(1+x^2\right)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is

(A) n^2y (B) $-n^2y$ (C) $-y$ (D) $2x^2y$

Numerical 13 questions

Q70. JEE Main 2025 (Online) 2nd April Morning Shift

Let $f : \mathbf{R} \to \mathbf{R}$ be a thrice differentiable odd function satisfying
$f'(x) \geq 0, f''(x) = f(x), f(0) = 0, f'(0) = 3$. Then $9f(\log_e 3)$ is equal to _____ .

Q71. JEE Main 2024 (Online) 4th April Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a thrice differentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeros of $(3f'f'' + ff''')(x)$ is _____.

Q72. JEE Main 2024 (Online) 1st February Evening Shift

If $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}\left(3\cos^2 x - 5\right)\cos^3 x$, then $96y'\left(\frac{\pi}{6}\right)$ is equal to :

Q73. JEE Main 2024 (Online) 27th January Morning Shift

Let $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3), x \in \mathbf{R}$. Then $f'(10)$ is equal to _____.

Q74. JEE Main 2023 (Online) 13th April Evening Shift

Let $f(x)=\sum_{k=1}^{10} kx^k, x \in \mathbb{R}$. If $2f(2)+f'(2)=119(2)^n+1$ then n is equal to _____

Q75. JEE Main 2023 (Online) 1st February Morning Shift

If $f(x)=x^2+g'(1)x+g''(2)$ and $g(x)=f(1)x^2+xf'(x)+f''(x)$, then the value of $f(4)-g(4)$ is equal to _____.

Q76. JEE Main 2023 (Online) 29th January Morning Shift

Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function that satisfies the relation $f(x+y)=f(x)+f(y)-1, \forall x,y \in \mathbb{R}$. If $f'(0)=2$, then $|f(-2)|$ is equal to _____.

Q77. JEE Main 2022 (Online) 27th July Evening Shift

For the curve $C:\left(x^2+y^2-3\right)+\left(x^2-y^2-1\right)^5=0$, the value of $3y'-y^3y''$, at the point (α,α), $\alpha>0$, on C, is equal to _____.

Q78. JEE Main 2022 (Online) 27th June Evening Shift

If $y(x)=(x^x)^x$, $x>0$, then $\frac{d^2x}{dy^2}+20$ at x = 1 is equal to _____.

Q79. JEE Main 2022 (Online) 26th June Evening Shift

Let f : R → R satisfy $f(x+y)=2^xf(y)+4^yf(x)$, ∀x, y ∈ R. If f(2) = 3, then $14.\frac{f'(4)}{f'(2)}$ is equal to _____.

Q80. JEE Main 2021 (Online) 26th August Morning Shift

If y = y(x) is an implicit function of x such that $\log_e(x+y)=4xy$, then $\frac{d^2y}{dx^2}$ at x = 0 is equal to _____.

Q81. JEE Main 2021 (Online) 17th March Morning Shift

If $f(x)=\sin\left(\cos^{-1}\left(\frac{1-2^{2x}}{1+2^{2x}}\right)\right)$ and its first derivative with respect to x is $-\frac{b}{a}\log_e 2$ when x = 1, where a and b are integers, then the minimum value of $|a^2-b^2|$ is _____ .

Q82. JEE Main 2020 (Online) 2nd September Evening Slot

If $y = \sum_{k=1}^{6} k\cos^{-1}\left\{\frac{3}{5}\cos kx - \frac{4}{5}\sin kx\right\}$,

then $\frac{dy}{dx}$ at x = 0 is _____.

Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **B**	3. **C**	4. **C**	5. **D**
6. **A**	7. **B**	8. **A**	9. **C**	10. *Scan QR*
11. **D**	12. **C**	13. **C**	14. **C**	15. **A**
16. **C**	17. **B**	18. **B**	19. **A**	20. **B**
21. **B**	22. **D**	23. **D**	24. **D**	25. **D**
26. **A**	27. **A**	28. **D**	29. **D**	30. **D**
31. **A**	32. **B**	33. **A**	34. **C**	35. **C**
36. **C**	37. **A**	38. **A**	39. **D**	40. **B**
41. **B**	42. **B**	43. **A**	44. **B**	45. **B**
46. **D**	47. **A**	48. **D**	49. **A**	50. **C**
51. **B**	52. **B**	53. **C**	54. **B**	55. **D**
56. **A**	57. **B**	58. **D**	59. **D**	60. **B**
61. **A**	62. **C**	63. **A**	64. **D**	65. **A**
66. **C**	67. **D**	68. **B**	69. **A**	

Numerical

70. **36**	71. **5**	72. **105**	73. **202**	74. **10**
75. **14**	76. **3**	77. **16**	78. **16**	79. **248**
80. **40**	81. **481**	82. **91**		